

Mathematics of Data Science - CA 1

Name: _____

Football Points Analysis

Consider two football teams, A and B. When they play against each other, the outcomes are as follows:

- Team A wins with probability 20%.
- The game is a draw with probability 40%.
- Team B wins with probability 40%.

The points are awarded as follows: a win is worth 3 points, a draw is 1 point, and a loss is 0 points. Let X be a random variable that counts the **total number of points Team A gets** in 10 mutually independent games against Team B.

1. Compute the expected value $\mathbb{E}[X]$.

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4 pts

2. Compute the variance $\text{Var}(X)$.

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3 pts

3. Let X_1 be the points for Team A in a single game and Y_1 be the points for Team B in that same game. Compute the covariance between their scores, $\text{Cov}(X_1, Y_1)$, and comment on its sign.

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3 pts

(a) First, we find the expected points from one game, $\mathbb{E}[X_i]$.

$$\mathbb{E}[X_i] = (3 \times 0.20) + (1 \times 0.40) + (0 \times 0.40) = 0.6 + 0.4 + 0 = 1$$

By the Linearity of Expectation, the expectation of the sum is the sum of expectations.

$$\mathbb{E}[X] = E \left[\sum_{i=1}^{10} X_i \right] = \sum_{i=1}^{10} \mathbb{E}[X_i] = 10 \times 1.0 = 10.$$

(b) To find the variance, we first need the variance of a single game, $\text{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2$. First, let's compute $\mathbb{E}[X_i^2]$:

$$\mathbb{E}[X_i^2] = (3^2 \times 0.20) + (1^2 \times 0.40) + (0^2 \times 0.40) = (9 \times 0.20) + (1 \times 0.40) + 0 = 1.8 + 0.4 = 2.2.$$

Now we can compute the variance of a single game:

$$\text{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = 2.2 - (1.0)^2 = 1.2.$$

Since the games are mutually independent, the variance of the sum is the sum of the variances.

$$\text{Var}(X) = \text{Var} \left(\sum_{i=1}^{10} X_i \right) = \sum_{i=1}^{10} \text{Var}(X_i) = 10 \times 1.2 = 12.$$

(c) The formula for covariance is $\text{Cov}(X_1, Y_1) = \mathbb{E}[X_1 Y_1] - \mathbb{E}[X_1] \mathbb{E}[Y_1]$. We already know $\mathbb{E}[X_1] = 1$. We compute also $\mathbb{E}[Y_1]$, the expected points for Team B. The distribution for Y_1 is:

- $\mathbb{P}(Y_1 = 3) \text{ (B wins)} = 0.40$
- $\mathbb{P}(Y_1 = 1) \text{ (Draw)} = 0.40$
- $\mathbb{P}(Y_1 = 0) \text{ (B loses)} = 0.20$

$$\mathbb{E}[Y_1] = (3 \times 0.40) + (1 \times 0.40) + (0 \times 0.20) = 1.2 + 0.4 + 0 = 1.6.$$

Next, we need $\mathbb{E}[X_1 Y_1]$. We must consider the joint outcomes for (X_1, Y_1) and apply LOTUS for the random vector (X_1, Y_1) :

- If A wins: $(X_1, Y_1) = (3, 0) \implies X_1 Y_1 = 0$. This has a probability of 0.20.
- If Draw: $(X_1, Y_1) = (1, 1) \implies X_1 Y_1 = 1$. This has a probability of 0.40.
- If B wins: $(X_1, Y_1) = (0, 3) \implies X_1 Y_1 = 0$. This has a probability of 0.40.

The expectation of the product is:

$$\mathbb{E}[X_1 Y_1] = (0 \times 0.20) + (1 \times 0.40) + (0 \times 0.40) = 0.4.$$

Finally, we compute the covariance:

$$\text{Cov}(X_1, Y_1) = \mathbb{E}[X_1 Y_1] - \mathbb{E}[X_1] \mathbb{E}[Y_1] = 0.4 - (1.0)(1.6) = 0.4 - 1.6 = -1.2.$$

The negative covariance indicates that as one team's score increases, the other's tends to decrease, which makes intuitive sense in a football match.